

In-Home Smartphone-Based Prediction of Obstructive Sleep Apnea in Conjunction With Level 2 Home Polysomnography

Seung Cheol Han, Daewoo Kim, Chae-Seo Rhee, Sung-Woo Cho, Vu Linh Le, Eun Sung Cho, Hyunggug Kim, In-Young Yoon, Hyeryung Jang, Joonki Hong, Dongheon Lee, Jeong-Whun Kim

[Author information](#)

[Article notes](#)

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This diagnostic study aims to validate the performance of a prediction model for obstructive sleep apnea using breathing sound recorded from smartphones in conjunction with level 2 polysomnography at home.

Key Points

The results of this study indicate the feasibility and accuracy of using deep-learning analysis of breathing sounds recorded by smartphones to potentially screen for sleep apnea in a home setting.

- **Question:** Can a sound analysis model effectively predict obstructive sleep apnea in a home environment, comparable with level 2 home polysomnography?
- **Findings:** This diagnostic study of 101 participants evaluated the performance of a deep-learning model analyzing breathing sounds recorded by smartphones at home in comparison with concurrent home polysomnography. The results demonstrated a high level of accuracy.
- **Meaning:** The results of this study indicate the feasibility and accuracy of using deep-learning analysis of breathing sounds recorded by smartphones to potentially screen for sleep apnea in a home setting.

Abstract

Importance

Consumer-level sleep analysis technologies have the potential to revolutionize the screening for obstructive sleep apnea (OSA). However, assessment of OSA prediction models based on in-home recording data is usually performed concurrently with level 1 in-laboratory polysomnography (PSG). Establishing the predictability of OSA using sound data recorded from smartphones based on level 2 PSG at home is important.

Objective

To validate the performance of a prediction model for OSA using breathing sound recorded from smartphones in conjunction with level 2 PSG at home.

Design, Setting, and Participants

This diagnostic study followed a prospective design, involving participants who underwent unattended level 2 home PSG. Breathing sounds were recorded during sleep using 2 smartphones, one with an iOS operating system and the other with an Android operating system, simultaneously with home PSG in participants' own home environment. Participants were 19 years and older, slept alone, and had either been diagnosed with OSA or had no previous diagnosis. The study was performed between February 2022 and February 2023.

Main Outcomes and Measures

Sensitivity, specificity, positive predictive value, negative predictive value, and accuracy of the predictive model based on the recorded breathing sounds.

Results

Of the 101 participants included during the study duration, the mean (SD) age was 48.3 (14.9) years, and 51 (50.5%) were female. For the iOS smartphone, the sensitivity values at apnea-hypopnea index (AHI) levels of 5, 15, and 30 per hour were 92.6%, 90.9%, and 93.3%, respectively, with specificities of 84.3%, 94.4%, and 94.4%, respectively. Similarly, for the Android smartphone, the sensitivity values at AHI levels of 5, 15, and 30 per hour were 92.2%, 90.0%, and 92.9%, respectively, with specificities of 84.0%, 94.4%, and 94.3%, respectively. The accuracy for the iOS smartphone was 88.6%, 93.3%, and 94.3%, respectively, and for the Android smartphone was 88.1%, 93.1%, and 94.1% at AHI levels of 5, 15, and 30 per hour, respectively.

Conclusions and Relevance

This diagnostic study demonstrated the feasibility of predicting OSA with a reasonable level of accuracy using breathing sounds obtained by smartphones during sleep at home.

Introduction

Obstructive sleep apnea (OSA) is characterized by recurrent episodes of upper airway obstruction during sleep and is associated with negative health consequences, including cardiovascular disease, stroke, and cognitive impairment.

[1,2,3](#)

Polysomnography (PSG) is the standard diagnostic method for OSA, but it is costly, inconvenient, and leads to underdiagnosis.

[4](#)

Moreover, the night-to-night variability of sleep patterns complicates diagnosis and monitoring.

[5,6](#)

To address these challenges, there is growing interest in developing home-based diagnostic approaches for OSA that are more accessible, affordable, and convenient. Home sleep apnea testing (HSAT) has been recommended by the American Academy of Sleep Medicine (AASM) as a suitable method for evaluating suspected OSA in adults.

[7](#)

However, HSAT is still relatively expensive and requires specialized equipment and trained personnel. Additionally, HSAT faces limitations in capturing the night-to-night variability of sleep, similar to PSG.

Sound-based assessment for OSA has emerged as a promising approach that enables noncontact and in-home monitoring using smartphones or wearable devices.

[8,9,10](#)

Previous studies have demonstrated the potential of using breathing sounds to develop predictive models for OSA detection. However, many of these studies validated their prediction models using sound collected in controlled environments, referencing level 1 PSG.

[8,10,11,12](#)

Furthermore, previous studies failed to account for home environment noises, which are essential for accurate OSA prediction in a home setting.

[10,12](#)

To address this, our predeveloped model was trained using data that incorporated various home noises, ensuring its ability to accurately predict respiratory events in such environments. However, it is important to note that the PSG data used in our previous studies were obtained in a sleep laboratory, which may not fully represent the home environment.

[11,13,14](#)

Therefore, the aim of this diagnostic study is to validate our prediction model by collecting smartphone sounds and home PSG data from participants' homes. By using home PSG results as the reference, this approach enables us to assess the performance of our prediction model under real-world conditions, encompassing diverse types of home noises.

Methods

Study Design

This is a diagnostic study that tried to validate the diagnostic accuracy of our predeveloped model that predicts the apnea-hypopnea index (AHI) from breathing sound. The study was performed between February 2022 and February 2023. The study was reviewed and approved by the institutional review board of Seoul National University Bundang Hospital. All participants provided written informed consent. The report generally followed the existing Standards for Reporting of Diagnostic Accuracy ([STARD](#)) 2015 protocol because STARD-AI has not been released yet.

Study Participants

The study included individuals 19 years and older who slept alone and had either been diagnosed with OSA or had no previous diagnosis. The exclusion criteria encompassed individuals with underlying neurological conditions such as cerebrovascular diseases, head trauma, and epilepsy, as well as psychiatric disorders and severe medical conditions, including kidney, liver, heart, and pulmonary diseases. The recruitment was performed by using bus, subway, and newspaper advertisements for research participants. In addition, patients who agreed to participate from the clinic of the Seoul National University Bundang Hospital otorhinolaryngology–head and neck department were also recruited.

Level 2 Full-Night Home PSG

All participants underwent unattended level 2 home PSG and smartphone sound recording. The home PSG was conducted using a portable instrument (Embletta MPR/ST+ Proxy [Natus Medical Inc]). From participants' homes, various physiological signals were recorded from electroencephalogram, electrooculogram, chin electromyogram, tibial electromyogram, electrocardiogram, respiratory inductance plethysmography, a nasal pressure transducer, an oral thermistor, a position sensor, and pulse oximetry. Participants visited the sleep laboratory to hook up portable instruments and returned home to carry out home PSG. Sleep technologists manually marked respiratory events. These annotations were then reviewed and confirmed by a sleep specialist based on the guidelines of the AASM scoring manual.

Smartphone-Based Collection of Breathing Sounds

Two smartphones, iPhone 13 (Apple Inc; using iOS operating system) and Galaxy S20 (Samsung Electronics; using Android operating system), were provided and placed next to the patient's bed, with the distance between the smartphones and the patient's head within the range of 50 to 100 cm ([Figure 1](#)). The setting of the smartphone at the participant's home and starting of the recording of breathing sound by application was all completed by the participants themselves by the guide paper the participants were provided. The collection of breathing sounds started before the individuals went to sleep and ended when they woke up in the morning. Our own recording application was developed to record the sound and collect data easily. The recording part of this application is exactly the same as default recording application in iOS and Android smartphones. The recording parameters included a sample rate of 16 000 Hz, a bit rate of 32 bits, and use of a single-channel microphone. We did not apply any audio filters.

Figure 1. Study Setting for Smartphone Recording and Level 2 Polysomnography (PSG) at Home.



Two smartphones, iPhone 13 (Apple Inc; using iOS operating system) and Galaxy S20 (Samsung Electronics; using Android operating system) were prepared near the individual's sleep side. The location was adjusted to make the distance between the smartphone and the patient's head 50 to 100 cm apart. The microphone (mic) at the lower part of the smartphone was directed to the patient's head.

Predeveloped Prediction Model

In this study, we used a predeveloped model that predicts the AHI from breathing sound. The model was built on a deep neural network architecture, and its detailed description can be found in a previous study.

Briefly, the prediction model was trained using a data set consisting of 1018 level 1 in-laboratory PSG (Embla N7000 [Natus Medical Inc]) audio data, 297 smartphone audio data recorded during level 1 in-laboratory PSG, and a total of 22 500 home noises to ensure its functionality in real-world home noise environments. The scoring of respiratory events, including apnea and hypopnea, was defined by the guidelines of the AASM scoring manual,

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both from the predeveloped model and also in this study. The cessation of airflow for at least 10 seconds was defined as apnea. On the other hand, a decrease in airflow more than 50% for at least 10 seconds or a moderate reduction in airflow for at least 10 seconds associated with arousals or oxygen desaturation ($<4\%$) were defined as hypopnea.

Statistical Analysis and Performance Matrix

We compared AHI predicted by 2 smartphones with AHI derived from home PSG using Pearson correlation analysis. Additionally, we evaluated the predictive performance of our model by using home PSG as the reference. The performance matrix included sensitivity, specificity, positive predictive value, negative predictive value, and accuracy. We considered different thresholds to divide participants into 2 classes based on home PSG AHI values (5, 15, and 30). Furthermore, we assessed the model's performance using 4-class

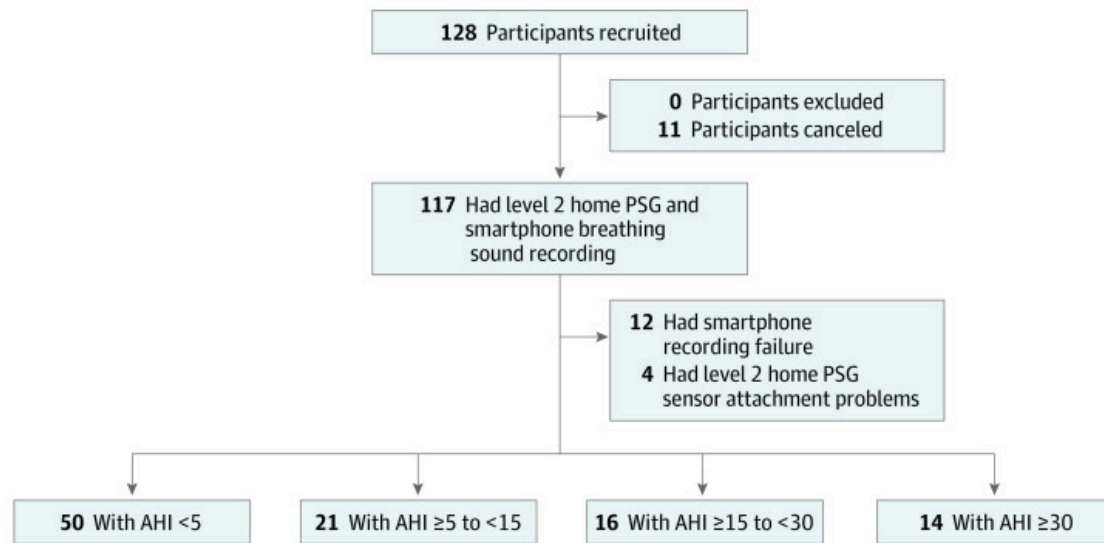
classification and evaluated accuracy and Cohen κ . The 4-class classification consisted of home PSG AHI values categorized as less than 5, between 5 and less than 15, between 15 and less than 30, and 30 or higher, representing the standard severity levels of OSA (normal, mild, moderate, and severe). All analyses were performed using IBM SPSS Statistics, version 26 (IBM Corp). Continuous parametric variables are presented as means (SDs).

Results

General Characteristics of Study Participants

A total of 128 participants were recruited, and 11 participants canceled before starting the study. Therefore, level 2 home PSG and smartphone breathing sound recording was performed in 117 participants. Among 4 participants, there were level 2 home PSG sensor attach errors. In addition, among 12 participants, there were smartphone recording failures due to smartphone device failure or participants' insufficient operation of recording application (eg, forgot to start the application) ([Figure 2](#)). A total of 101 participants had complete data from their smartphones and home PSG. There were 50 male participants and 51 female participants. The age of the participants ranged from 22 to 79 years, with a mean (SD) age of 48.3 (14.9) years. All of the participants were Korean. The participants had a mean (SD) body mass index (calculated as weight in kilograms divided by height in meters squared) of 24.5 (3.8). eTable 1 in [Supplement 1](#) summarizes the general characteristics of normal (n = 50; 16 male participants), mild (n = 21; 10 male participants), moderate (n = 16; 13 male participants), and severe (n = 14; 11 male participants) OSA subgroups; the mean (SD) ages for each subgroup were 44.5 (15.4), 52.7 (14.9), 54.0 (11.6), and 49.5 (14.4) years, respectively, and the mean (SD) body mass index was 23.4 (3.9), 24.9 (3.0), 25.4 (4.1), and 26.9 (3.9), respectively.

Figure 2. Flowchart of Participants Recruited for the Study.



A total of 128 participants were recruited. There were no participants excluded, but 11 participants canceled before starting the study. Level 2 home polysomnography (PSG) and smartphone breathing sound recording was performed in 117 participants. Four participants had level 2 home PSG sensor attach errors, and 12 participants had smartphone recording failures. There were 50 participants with a normal severity level of obstructive sleep apnea (apnea-hypopnea index [AHI] <5), 21 with mild (AHI ≥5 to <15), 16 with moderate (AHI ≥15 to <30), and 14 with severe (AHI ≥30).

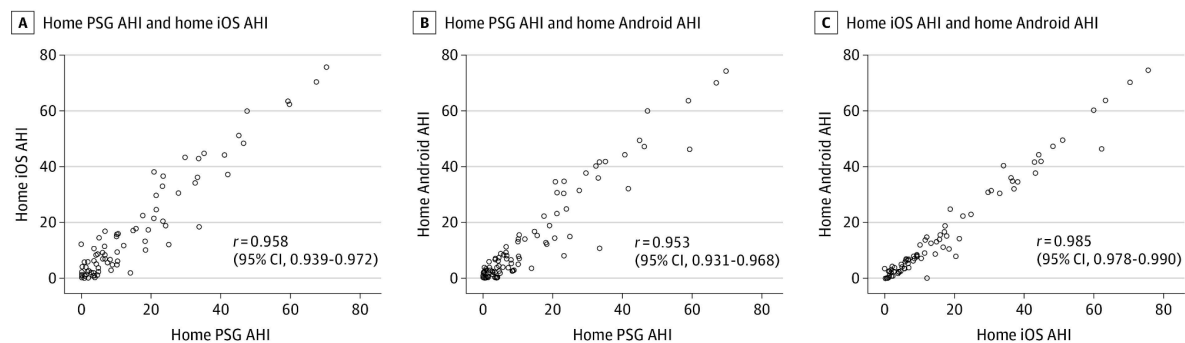
AHI From Level 2 Home PSG and Smartphones

AHI derived from home PSG ranged from 0 to 69.7 per hour, with a mean (SD) AHI of 12.4 (16.0) per hour. In each subgroup categorized by OSA severity, the mean (SD) AHI from home PSG was 1.7 (1.7), 8.9 (2.6), 21.7 (3.7), and 45.9 (13.0) per hour for normal, mild, moderate, and severe, respectively. AHI derived from the home iOS smartphone ranged from 0 to 75.6 per hour, with a mean (SD) AHI of 13.7 (17.8) per hour. In each subgroup based on home PSG, the mean (SD) AHI from the home iOS smartphone was 2.2 (2.8), 9.4 (4.8), 24.4 (9.9), and 49.2 (15.7) per hour for normal, mild, moderate, and severe, respectively. Similarly, AHI derived from the home Android smartphone ranged from 0 to 74.0 per hour, with a mean AHI of 12.6 (17.2) per hour. In each subgroup based on home PSG, the mean AHI from the home Android smartphone was 1.7 (2.1), 8.2 (4.3), 22.7 (9.5), and 46.8 (16.4) per hour for normal, mild, moderate, and severe, respectively (eTable 1 in [Supplement 1](#)).

Correlations Between Home PSG AHI and Home Smartphone AHI

The correlations between the different modalities are presented in [Figure 3](#). AHI derived from home PSG exhibited a strong correlation with AHI from the home iOS smartphone ($r = 0.958$; 95% CI, 0.939-0.972; [Figure 3A](#)). Additionally, AHI from home PSG showed a strong correlation with AHI from the home Android smartphone ($r = 0.953$; 95% CI, 0.931-0.968; [Figure 3B](#)). Furthermore, there was a strong correlation between AHI from the home iOS smartphone and AHI from the home Android smartphone ($r = 0.985$; 95% CI, 0.978-0.990; [Figure 3C](#)).

Figure 3. Correlation Between Home Polysomnography (PSG) Apnea-Hypopnea Index (AHI) and Home Smartphone AHI Predicted From Breathing Sound Recorded.



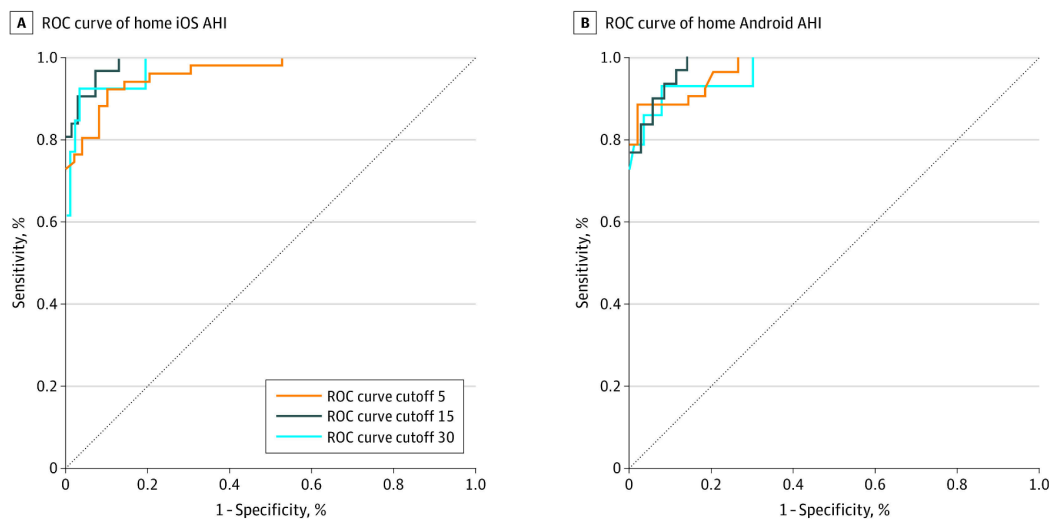
Participants used either iPhone 13 (Apple Inc; using iOS operating system) or Galaxy S20 (Samsung Electronics; using Android operating system) smartphones for recording. AHI predicted by the 2 smartphones was compared with AHI derived from home PSG using Pearson correlation analysis.

Performance of Binary Classification for OSA

The performance of binary classification using AHI cutoff values of 5, 15, and 30 per hour is described by receiver operating characteristic curves in [Figure 4](#). For the iOS smartphone, the sensitivity at AHI cutoffs of 5, 15, and 30 per hour

was 92.6%, 90.9%, and 93.3%, and the specificity was 84.3%, 94.4%, and 94.4%, respectively. The accuracy of the iOS smartphone was 88.6%, 93.3%, and 94.3% at AHI cutoffs of 5, 15, and 30 per hour, respectively.

Figure 4. Performance of Binary Classification for Home Smartphone Apnea-Hypopnea Index (AHI) According to the AHI Cutoff Values.



Participants used either iPhone 13 (Apple Inc; using iOS operating system) or Galaxy S20 (Samsung Electronics; using Android operating system) smartphones for recording. ROC indicates receiver operating characteristic.

For the Android smartphone, the sensitivity at AHI cutoffs of 5, 15, and 30 per hour was 92.2%, 90.0%, and 92.9%, and the specificity was 84.0%, 94.4%, and 94.3%, respectively. The accuracy of the Android smartphone was 88.1%, 93.1%, and 94.1% at AHI cutoffs of 5, 15, and 30 per hour, respectively. These findings can also be found in eTables 2 and 3 in [Supplement 1](#).

Performance of 4-Class Classification for OSA

In the 4-class classification, participants were categorized into 4 subgroups according to their home PSG AHI values: normal, mild, moderate, and severe.

The accuracy for both iOS and Android smartphones was 75.2%, with a Cohen κ of 0.636 and 0.627, respectively ([Table](#)).

Table. Classification Performance According to the Number of Classes.

Classification ^a	% (95% CI) ^b	
	Home iOS	Home Android
4 Class		
Accuracy	75.2 (66.3-84.2)	75.2 (66.3-83.2)
Cohen κ	0.636 (0.518-0.755)	0.627 (0.499-0.746)
2 Class		
AHI cutoff of 5		
Accuracy	88.6 (82.2-95.0)	88.1 (83.2-95.0)
Cohen κ	0.762 (0.630-0.898)	0.782 (0.660-0.900)
AHI cutoff of 15		
Accuracy	93.3 (88.1-97.0)	93.1 (87.1-97.0)
Cohen κ	0.836 (0.712-0.935)	0.803 (0.654-0.923)
AHI cutoff of 30		
Accuracy	94.3 (89.1-98.0)	94.1 (88.1-97.0)
Cohen κ	0.778 (0.587-0.926)	0.748 (0.539-0.905)

a Apnea-hypopnea index less than 5 indicates a normal level of obstructive sleep apnea; mild, between 5 and less than 15; moderate, between 15 and less than 30; and severe, 30 or higher.

b Participants used either iPhone 13 (Apple Inc; using iOS operating system) or Galaxy S20 (Samsung Electronics; using Android operating system) smartphones for recording home apnea-hypopnea index values.

Discussion

This study provides validation for home smartphone AHI, using level 2 full-night home PSG as a reference. This study is also one of the pioneering studies in collecting real-world data rather than relying on in-laboratory data sets. Previously, to our knowledge, only 1 study used smartphone-recorded

sounds in a home setting, referencing level 3 HSAT (SOMNOtouch RESP [SOMNOmedics GmbH]).

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Therefore, no electroencephalogram was incorporated, potentially resulting in an underestimation of AHI. Furthermore, the prediction model was developed using only 103 participants. The training and test data were obtained from the same participants, and the model was not trained for various noises, while the present model was trained using 22 500 home environment noises.

This validation study was conducted at home and demonstrated high performances in predicting AHI using iPhone 13 and Galaxy S20; both predictions exhibited a strong correlation with home PSG AHI, with correlation coefficients of 0.958 and 0.953, respectively. Furthermore, there was a high correlation between the results obtained from 2 different smartphone operating systems, with a correlation coefficient of 0.985. These findings highlight a robust correlation between AHI predicted by smartphones and the results obtained from home PSG, regardless of the smartphone's operating system. Moreover, we evaluated the performance of AHI prediction through binary classification with home PSG AHI cutoff values of 5, 15, and 30 per hour, as well as through 4-class classifications. The Cohen κ values, indicating the agreement between predicted and reference values, fell within the substantial performance range (typically considered to be between 0.6 and 0.8) for both iOS and Android smartphones in the 2-class and 4-class AHI classifications. We especially want to highlight the good performance of binary classification using AHI cutoff values of 5 and 15 per hour. These values have potentially meaningful implications for screening, both in determining OSA vs no OSA and in screening for more severe OSA with long-term adverse health consequences.

Various devices have been developed for predicting OSA, including wearables and nearables. Wearables, such as smartwatches (eg, Apple Watch Series 8, Galaxy Watch5, Google Pixel Watch, Fitbit Sense 2) and ring-shaped devices (eg, Oura Ring 3), have gained popularity in the market.

[16,17,18,19,20](#)

These devices typically use sensors like photoplethysmography and accelerometers or gyroscopes to track heart rate variability and movement. However, a limitation of wearables is their primary focus on measuring oxygen desaturation rather than accurately diagnosing OSA based on AHI. While they may detect desaturation events, their reliability in accurately determining AHI is questionable. In contrast, the prediction model in this study offers considerable advantages over wearables because it can precisely calculate AHI on an epoch-by-epoch basis and is noncontact, avoiding any sleep disruption.

On the other hand, nearables are positioned in close proximity to the user's body and use noncontact methods to track sleep. These devices use sensors like radar or piezoelectric pads placed under the mattress to detect subtle movements related to body movement, breathing, and heart rate during sleep.

[21,22](#)

A notable device in this field is the Withings Sleep analyzer (WSA [Withings]), which can be conveniently placed under the mattress without the need for a technician. The WSA uses a deep-learning algorithm that incorporates body movement, breathing pattern, cardiac activity, and snoring to calculate AHI. A recent study evaluating the performance of the WSA involved 118 patients from 2 centers.

However, it is important to note that this study solely compared the results between level 1 PSG and WSA for a single night within a controlled sleep laboratory setting, without considering real-world sounds or noise, unlike the present study. Furthermore, in terms of binary AHI threshold classification, the WSA exhibited a sensitivity of 88% and specificity of 88.6% for an AHI threshold of 15 and a sensitivity of 86% and specificity of 91.2% for an AHI threshold of 30. Both outcomes fall behind the performance of our model. These results are intriguing, especially considering that the study evaluating the WSA was conducted in a controlled sleep laboratory environment, which differs from a real home setting.

Numerous studies have been conducted to predict or screen for OSA using various types of sounds produced by patients during sleep.

[24,25,26,27,28](#)

Most of these studies use smartphones or recording devices, aiming to create airable devices for OSA screening or diagnosis. Airable devices take advantage of widely used electronic devices like smartphones that are integrated into people's daily lives. Typically implemented as applications, these devices can be easily installed on personal mobile devices, ensuring convenient access for patients. Some studies focus on tracheal sounds,

[24,25,26](#)

while others use speech

or ambient sound.

[8,9,27](#)

However, many of these studies conducted sound recordings in controlled, noise-free environments,

[26,27,28](#)

which do not accurately represent the typical background noises experienced by patients in their home sleep settings. The ultimate goal of using sound-based approaches for OSA prediction is to develop a method that can be applied in a home setting, similar to level 1 in-laboratory PSG. Although some studies have recorded sounds in a home environment, they often neglect the presence of ambient noise.

[24,25](#)

These studies demonstrate the potential for handling less severe noise disturbances in a home setting, but they require patients to attach smartphones or microphones to their bodies during sleep, making them contact devices rather than airable devices like other studies. Only a few studies have taken into account the presence of real-life ambient noise in the sleeping environment. For example, a study used level 1 full-night PSG along with a noncontact digital voice recorder.

In that study, the level 1 PSG was conducted in a regular room without noise attenuation equipment to create a more realistic setting. However, it should be noted that all participants underwent the test in the same room within the hospital, which limits the generalizability to diverse real-world situations despite considering noise. One notable advantage of our prediction model is that it explicitly considers noises in a home setting by incorporating it during the development of the model. This allowed us to use breathing sounds recorded by smartphones placed at a distance from the patients' beds in the home setting. Therefore, we believe that the inclusion of ambient noises enhanced the real-world applicability of our model.

This study specifically focused on predicting OSA in a home-based setting, which has received relatively less attention. Many studies about the prediction of OSA usually used level 1 in-laboratory PSG as the standard for comparison.

[8,24,25](#)

In our previous work, we developed and validated a deep-learning model using in-laboratory PSG data.

[14](#)

For analyzing patients' sleep on a night-by-night basis, we validated the model's performance based on level 2 home PSG in the present study.

Relying solely on in-laboratory PSG may be insufficient for comprehensive assessment of a patient's sleep patterns due to several reasons. Level 1 in-laboratory PSG cannot capture night-to-night variations. Also, it fails to replicate the various factors present in the patient's home sleep environment.

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The sleep patterns of individuals vary based on their chronotype, lifestyle, and home environment, including differences in the time they go to bed. However, in the sleep laboratory where level 1 PSG is typically conducted, all individuals are required to go to sleep at the same time and in the same environment. This can affect the quality of sleep during the test.

Therefore, obtaining multnight, home-setting PSG results is crucial to accurately evaluate a patient's sleep. Our approach will allow for the measurement of these variabilities. Additionally, our prediction model demonstrated a reasonable level of agreement with level 2 home PSG. Our deep-learning model, which predicts AHI using breathing sounds recorded by smartphones, can assist clinicians in diagnosing and assessing patients with OSA in their real-world environments, in conjunction with PSG data. We also think that it will be a valuable tool especially in cases where there are discrepancies between the patient's reported history and the results of in-laboratory PSG.

Limitations

There are several limitations that should be acknowledged in this study. First, our focus was on the total AHI, and we did not investigate changes in AHI in relation to different positions of the patients. Since we relied solely on breathing sounds captured by smartphones, we were unable to gather information

regarding the patients' positional changes or oxygen saturation data. This restricts our ability to accurately analyze the effects of different positions on AHI or oxygen saturation change during sleep. Future studies will be conducted to predict positions based on breathing sound and could also consider incorporating position sensors or oximetry sensors. Second, it is important to note that this study included only Korean participants and a small validation population. Therefore, the generalizability of our prediction model to other racial and ethnic groups may be limited. To ensure the global applicability of our model, further research involving a larger population of participants and participants from diverse racial and ethnic backgrounds would be necessary. Third, Cohen κ of 4-class classification was more than 0.6, which is substantial, but the model's performance is not as good as the binary classification model's performance. We are trying to advance our model to enhance 4-class's performance also. Fourth, level 2 home PSG results notoriously underestimate OSA and AHI levels compared with level 1 in-laboratory PSG results. Even though we demonstrated high accuracy to a level 2 home PSG in this study, the model may still be underestimating OSA and AHI levels. Despite these limitations, our study represents an important advancement in the validation and application of a deep-learning model for predicting OSA using breathing sounds recorded by smartphones.

Conclusions

This diagnostic study has produced promising results regarding the performance of smartphone-based prediction of AHI in a home setting. The validation of smartphone AHI with level 2 home PSG as a reference suggests that a future scenario in which clinicians use both in-laboratory PSG and home smartphones to comprehensively assess the sleep characteristics of patients with OSA over multiple nights is within reach. By harnessing the convenience and accessibility of smartphones, our prediction model offers the potential for a more holistic understanding in individuals' sleep patterns beyond the limitations of the sleep laboratory. However, it is important to acknowledge that continuous refinement and validation of the model will be critical to ensure the accuracy and reliability of the results.

